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Depression Detection of Reddit Comments

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by

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ABSTRACT OF THE THESIS

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As mental health awareness increases, there has been growing interest in developing machine learning models to assist with early detection of depression. Building on prior studies that have explored this task, this study aims to replicate and extend previous findings using a different dataset and models. Approximately 10,000 comments were collected from each of the subreddits r/depression and r/AskReddit, resulting in a dataset of roughly 20,000 Reddit comments. The task is a binary classification problem to detect depressive language and to compare the performance of classical machine learning models with modern deep learning models. The classical models—Logistic Regression, k-Nearest Neighbors, and Multinomial Naive Bayes, were evaluated alongside a feedforward neural network and a fine-tuned BERT model. Results show that BERT achieved the highest performance at approximately 85.2% accuracy and 86.2% recall for the depressed class. However, Multinomial Naive Bayes had the second best performance with roughly 79.8% accuracy and 88.7% recall for the depressed class despite its simplicity. These findings suggest that in some applications, it may be advantageous to accept a modest reduction in predictive performance in exchange for substantially lower computational costs and model complexity.

The thesis of Clayton Wai-Kai Chan is approved.

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2026

*To my advisor who has not only been my professor,
but a personal mentor, life coach, and uncle-figure...*

Waffles Whine & Cheese

*I'm also grateful for every new connection I made at UCLA
and couldn't have done this without them.*

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CHAPTER 1

Introduction

Depression is one of the most prevalent mental health conditions worldwide, and awareness of mental health issues has increased substantially in recent years. Alongside this increased awareness, social media platforms have become spaces where individuals openly discuss their thoughts, emotions, and personal struggles. As a result, social media data has attracted growing attention from researchers interested in developing methods for the early detection of mental health conditions.

While much of the existing literature explores a wide range of deep learning models, relatively fewer classical machine learning methods have been examined in depth. Consequently, this study aims to replicate aspects of prior research while extending the analysis by evaluating a broader set of classical machine learning algorithms. By doing so, this work seeks to determine whether simpler models can achieve comparable performance while substantially lowering computational resources.

Accordingly, this thesis addresses the following research questions:

1. Which model performs best in terms of True Positive Rate and accuracy when detecting depression from Reddit text?
2. How large is the performance gap between the best classical machine learning method and modern deep learning approaches?
3. Is the difference in performance substantial enough to justify the additional computational costs associated with deep learning models?

CHAPTER 2

Literature Review

The primary inspiration for this paper comes from a study conducted by a researcher at Stanford [1]. In that study, the author collected comments from r/AskReddit and r/depression and a binary classification task was performed to detect signs of depression in the comments. The authors evaluated several deep learning models, including BERT and convolutional neural networks (CNNs), while logistic regression and support vector machines (SVMs) were used as classical baseline models. Their results showed that the CNN achieved the best performance, with an accuracy of 92.5% and a recall of 95.6% for the depressed class. In comparison, the best-performing classical model, SVM, achieved an accuracy of 85% and a recall of 82.1%.

Similarly, another study analyzed tweets from Twitter to perform depression detection using a comparable binary classification framework [2]. This study compared several classical machine learning models—logistic regression, Bernoulli Naive Bayes, and random forests, with transformer-based models derived from BERT. Among the models tested, RoBERTa achieved the highest performance, with an accuracy of 98% and a recall of 99%. However, logistic regression performed nearly as well, achieving both accuracy and recall values of approximately 97%.

Furthermore, the motivation for investigating whether computational costs can be reduced is partly inspired by another study that performed multiclass classification on text data related to various mental health conditions [3]. In this work, classical models—including logistic regression, SVM, and random forests—were compared with a large language model

(LLM). The study raised an important question regarding whether the performance improvements provided by LLMs justify their significantly greater computational cost. The author found that the LLM achieved an accuracy of 86%, while the best-performing classical model, SVM, achieved an accuracy of 76%. Although the LLM outperformed the classical models, the results highlighted that traditional approaches can still perform reasonably well despite requiring far fewer computational resources.

Building on these studies, this paper combines elements from each approach to conduct a comparative analysis. Similar to the first study, we construct a dataset consisting of 10,000 comments each from r/depression and r/AskReddit. We then compare classical machine learning models with deep learning models to evaluate their performance on the depression detection task. Our classical models include logistic regression with L_2 regularization, Multinomial Naive Bayes, and k-nearest neighbors (k-NN). Multinomial Naive Bayes and k-NN were selected due to their frequent use in natural language processing tasks and lack of appearance in the literature. For the deep learning models, we utilize a fine-tuned BERT model inspired by prior studies, while also exploring a simpler feed-forward neural network architecture. Many prior studies have focused on more complex neural architectures; therefore, including a simpler neural network provides an additional point of comparison. Through this analysis, we aim to further investigate whether comparable performance can be achieved with lower computational cost, particularly when using classical machine learning models.

CHAPTER 3

Data

3.1 Data Collection

The dataset used in this study was originally sourced from an anonymous developer, and the corresponding repository can be found in [4]. The original dataset pulled Reddit comments spanning from 2005 to 2024 with an API, stored in compressed .zst files. From this archive, all comments from the subreddits r/AskReddit and r/depression were extracted for analysis.

Because the full dataset covers nearly two decades of Reddit activity, the extracted files were extremely large. The r/depression dataset was approximately 1 GB in size, while the r/AskReddit dataset was approximately 90 GB. These sizes would increase further if the data were converted into a .csv format for analysis.

Due to the limited computational resources available for this study, as well as the presence of a substantial class imbalance, a subset of the data was used. Specifically, 10,000 comments were randomly sampled from each subreddit. This approach provided a manageable dataset size while maintaining a balanced class distribution, thereby reducing the likelihood that the models would default to predicting the majority class (i.e., classifying all comments as non-depressed).

3.2 Data Ethics & Privatization

An important part of research is following some ethical codes. Due to the sensitive nature of mental health data, care was taken to privatize and anonymize the comments of identifiable information. The original data set had columns that contained the metadata of the comments such as the timestamp or upvotes. Such irrelevant columns were subsequently deleted and only one column was kept, which was the body or actual comment itself. Furthermore, regular expressions (regex) were used to redact parts of the comment that contained sensitive information such as the user. Thus, our current codebook looks like

Name	Description
body	The text/body of the Reddit comment
is_depressed	0 for control/not depressed (from r/AskReddit) and 1 for depressed (from r/depression)

Table 3.1: Code Book

As a disclaimer, the data was used for nonprofit research purposes only.

3.3 Data Pre-processing

Depending on the model in question, the data was pre-processed in tiers.

3.3.1 Missing values

A lot of observations had bodies that read as [deleted] , “”, or NaN. These observations were simply removed as it is difficult to impute text data. As a result, we are left with ~18,000 observations.

3.3.2 English Comments

Since several hundred comments out of the 20,000 were not written in English, and the analysis focuses on features derived from the English language, these comments were removed from the dataset. Removing non-English comments simplified the dataset and reduced unnecessary complexity for the models. This step marks the end of pre-processing for the BERT model, as it is generally recommended to keep the input text largely intact when using transformer-based models.

3.3.3 Standardization

In order to reduce redundancy to improve model performance, the text was standardized in the following steps.

- Typos were fixed
- Contractions were removed e.g. should've → should have
- Text was lowercased
- Removal of numbers, punctuation, and special characters
- Removal of stopwords to reduce noise
- Removal of gibberish not in the standard English dictionary
- Words were lemmatized to root words e.g. confused, confusing → confuse

3.4 Feature Extraction

3.4.1 Term Frequency

The next big step is to convert the text into a data frame with features that can be used for analysis. This was done using CountVectorizer or a Bag-of-Words model which compiled all the words that appeared in every comment/document and counted the term frequency of each word to build our data frame. The term frequency is defined as:

$$\text{TF}(t, d) = \text{Raw Count of term } t \text{ appearing in document } d$$

where it is the number of times the term t appears in a document d . The final data frame had approximately 11,000+ unique words/columns and 18,000+ documents/observations with the values being term frequency. It is important to note the loss of complexity in word order and context, which may result in a decline in the performance of the models. However, since we are more interested in the words that signal depression, losing word order is not a big issue. The pre-processing stops here for Multinomial Naive Bayes as this is a discrete model.

3.4.2 Term Frequency-Inverse Document Frequency (TF-IDF) Transformation

For the rest of the models, a TF-IDF transformation was applied to provide more equity for the weighting of the terms which may improve model performance. The TF-IDF transformation is as follows:

$$\text{TF} * \text{IDF}(t, d, D) = \text{TF}(t, d) \times \log \frac{N}{1 + k}$$

where N is the total number of documents and $k = |\{d \in D : t \in d\}|$ is the number of documents containing the term t and D is the set of all documents with d representing an element of the document set. High frequency words were down-weighted to avoid their dominance while rare words were weighted up as they may be important words. The rows were then normalized by the L_2 norm.

3.5 Cross Validation

80% of the data was used to train the model while 20% of the data was used to test the performance of the model. However, hyperparameter tuning used cross validation on 5 folds to prevent overfitting. The output variable `is_depressed` was also stratified to ensure there were equal observations of depressed and not depressed.

CHAPTER 4

Exploratory Data Analysis

Before building models, it is useful to first explore the data to identify any potential patterns. Since the dataset consists of text, we will focus on examining word frequencies using word clouds and frequency plots. In addition, we will compare the most frequent words between the depressed and non-depressed classes to observe any notable differences in language use.

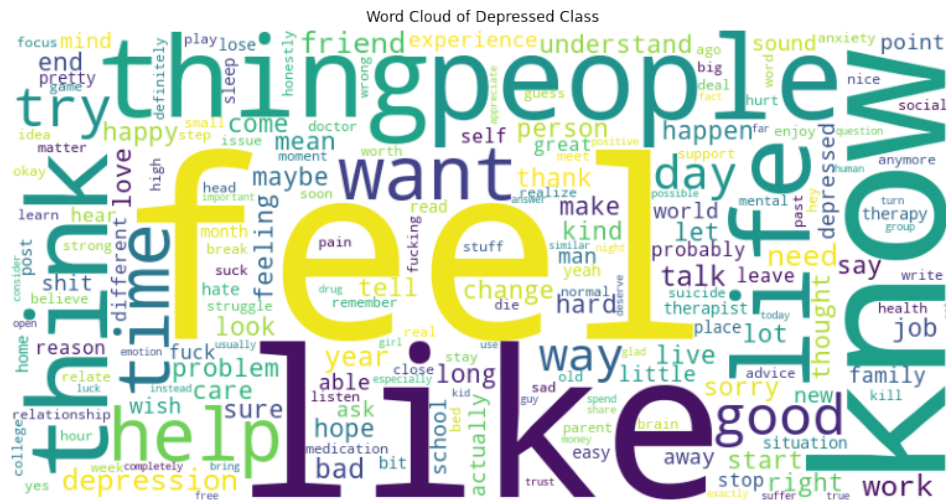


Figure 4.1: Word Cloud of Depressed Class

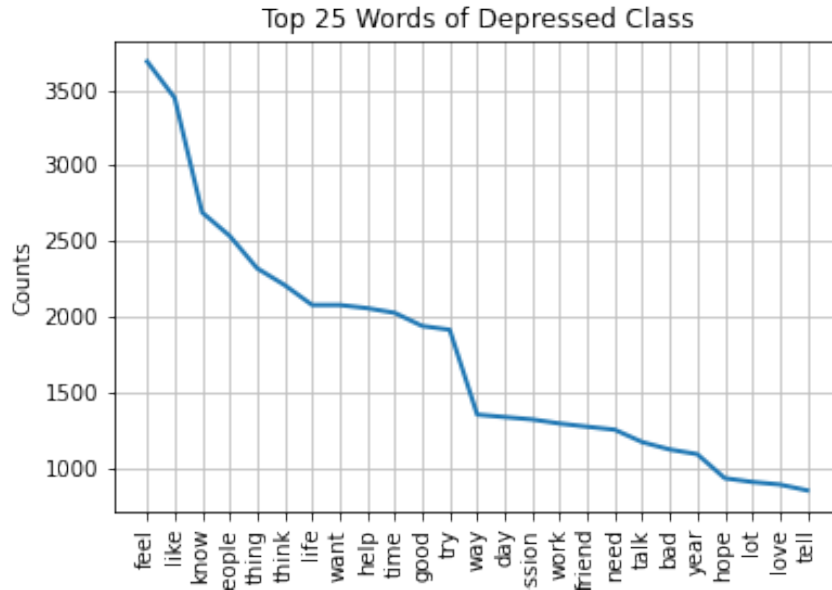


Figure 4.2: Top 25 Most Frequent Words of Depressed Class

We begin by examining the most common words that are unique to each class. For the depressed class, the most frequent unique words were [“depression”, “help”, “talk”, “hope”, “love”]. It is not surprising that “depression” appears among the most frequent words in this category. Additionally, words such as “help” and “talk” occur frequently, which may reflect discussions about seeking support or professional assistance, such as speaking with a therapist.

On the other hand, the most frequent words unique to the non-depressed class were [“say”, “look”, “come”, “guy”, “night”]. What is notable is that both classes share 20 words in common among their top 25 most frequent words. However, these shared words appear significantly more often in the depressed class, roughly two to three times as frequently as in the non-depressed class. For example, the word “feel” appeared 1,284 times in the non-depressed class, whereas it appeared 3,689 times in the depressed class. This difference is intuitive, as posts in the depression subreddit are more likely to involve candid discussions about personal emotions and experiences.

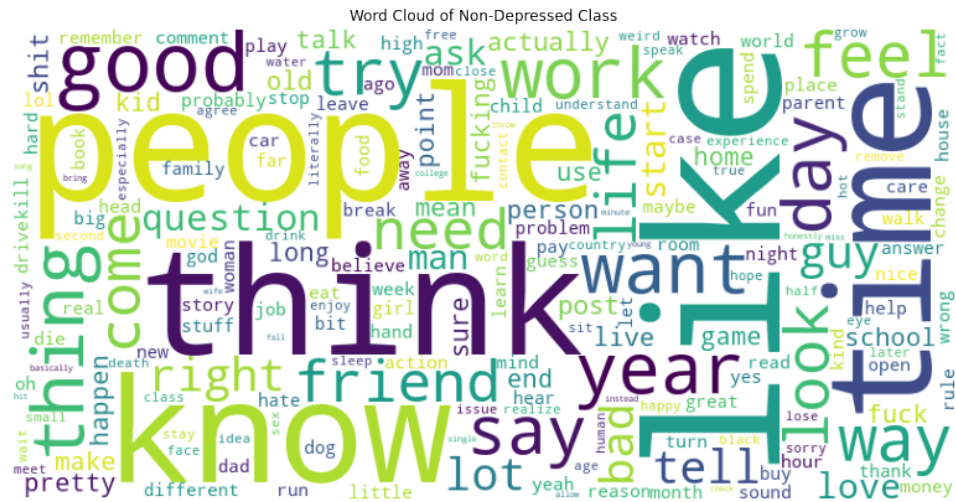


Figure 4.3: Word Cloud of Non-Depressed Class

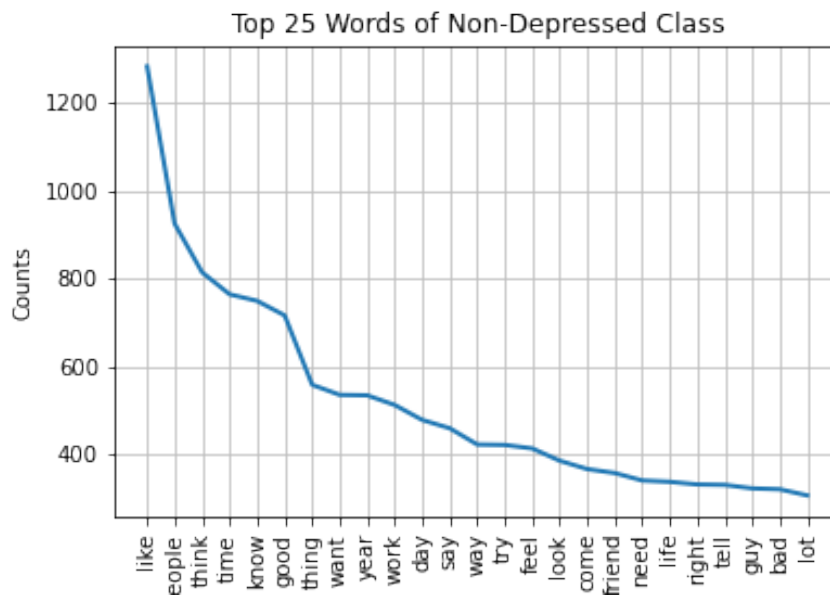


Figure 4.4: Top 25 Most Frequent Words of Non-Depressed Class

CHAPTER 5

Methodology

As stated earlier, the goal of this study is to compare classical models with deep learning models to determine whether computational costs could be reduced. Logistic regression was selected as a baseline for comparison. In addition, conventional models commonly used for Natural Language Processing (NLP)—k-nearest neighbors (k-NN) and Multinomial Naive Bayes, were included. A feed-forward neural network was also used as a baseline for the deep learning approaches. Furthermore, because the task involves text data, a large language model was incorporated, specifically a fine-tuned Bidirectional Encoder Representations from Transformers (BERT) model, to better capture contextual information and nuances that earlier models may fail to detect.

5.1 Classical Machine Learning Models

5.1.1 L_2 Logistic Regression

Due to its omnipresence in the literature, logistic regression will serve as a baseline. Since the data has many features, an L_2 penalty will be added to prevent overfitting and shrink the coefficients of the variables that are not useful for predictions and only add noise. Thus, our optimization function will look like the following:

$$\operatorname{argmin}_{\boldsymbol{\beta}} \left[- \sum_{i=1}^n (y_i \log(p_i) + (1 - y_i) \log(1 - p_i)) + \lambda \|\boldsymbol{\beta}\|_2^2 \right]$$

where the coefficients are chosen that minimizes the negative log-likelihood and squared 2-norm of the coefficients β with λ as the penalty term to control overfitting and output variable is depressed $y_i \in \{0, 1\}$.

With cross validation, the best λ was found to be 0.875. Furthermore, given the input vector $\mathbf{x}_i^T$, the predicted probability is p_i defined as

$$p_i = \frac{1}{1 + e^{-\mathbf{x}_i^T \hat{\beta}}}$$

where $\hat{\beta}$ are the estimated coefficients and $p_i > 0.5$ is classified as depressed.

5.1.2 k-Nearest Neighbors

Another common model for NLP tasks is k-NN. However, instead of Euclidean distance, cosine distance is used to measure word similarity. Since the data is normalized through the TF-IDF transformation, we are left with unit vectors. Hence, when calculating the cosine distances between two observations/vectors $\vec{a}, \vec{b}$,

$$\cos(\theta) = \vec{a} \cdot \vec{b}$$

where the data is classified based on the majority class of the top k values with the largest cosine distances. Regarding k , a grid search was performed to find the k that optimized the accuracy. Values up to $\sqrt{n} \approx 119$, where n was the number of training samples was searched; only odd values were considered to prevent ties. The optimal k was found to be 15.

5.1.3 Multinomial Naive Bayes

Given that the data is discrete count data, Multinomial Naive Bayes would be an apt model.

The model is as follows:

$$\begin{aligned}\text{Posterior} &= \text{Prior} \times \text{Likelihood} \\ P(C|X_1, \dots, X_n) &\propto P(C)P(X_1, \dots, X_n|C) \\ &\propto P(C) \prod_{i=1}^n P(X_i|C) \\ &\propto P(C) \prod_{i=1}^n p_i^{n_i}\end{aligned}$$

where $P(C)$ is the proportion of class C in the training set, $p_i = P(X_i|C) = \frac{\text{Count}(X_i, C) + 1}{N + V}$, and n_i is the number of times X_i appears. Using the training data, $\text{Count}(X_i, C)$ is the number of times the word X_i appears in class C , N is the total amount of words in class C , and V is vocabulary, the number of unique words across all documents, with 1 being there to prevent 0 probabilities. Once the posterior probabilities are calculated, the observation is classified as the class with the highest posterior probability. The model is intuitive as the higher the frequency of a word in a certain class, the higher the posterior probability of being in that class if observed in the data.

5.2 Deep Learning

5.2.1 Feed-Forward Neural Network

A step up from classical models would be to look at more complex deep learning models. The complexity comes from the architecture and the amount of hyperparameters that can go into a neural network. Hence, parallel computing was used to perform a grid search for various combinations of hyperparameters. The optimal model was found to have the following architecture and hyperparameters.

We have already embedded our text with a TF-IDF transformation, so we will feed it as our input layer X with 11,713 neurons representing the vocabulary size of the data. Next, we have a single hidden layer with 128 neurons and a ReLU activation function. The hidden layer h is obtained by multiplying the weight matrix W with X and adding some biases b . Finally, there is a single output neuron to give us a logit score and a sigmoid activation function to give us a probability p for binary classification. This is obtained by the unembed weight matrix W_u and biases b_u . Thus, the neural network is defined to be:

$$h = \text{ReLU}(WX + b)$$

$$p = \sigma(W_u h + b_u)$$

Training details include using a Binary Cross Entropy Loss with an Adam Optimizer. A learning rate of 0.01 was used along with 50 epochs. Finally, an $\alpha = 0.001$ was used as the L_2 penalty to prevent overfitting.

5.2.2 Bidirectional Encoder Representations from Transformers (BERT)

The last model discussed will be the most complex, which is a transformer neural network, BERT. For the BERT model, we did not train from scratch and fine-tuned it instead. The architecture of the fine-tuned BERT model starts with the input layer. First, the model must transform the text into numbers, which is achieved by creating word embeddings with embed matrix W_e multiplied with input x . Additionally, positional embeddings from the pre-trained model are added to keep track of word order. This is something all of the models up to this point could not do. Thus the first hidden layer h_1 is created by adding the positional embeddings and word embeddings defined in the following equations:

$$h_1 = \text{Word Embedding} + \text{Positional Embedding}$$
$$\text{Word Embedding} = W_e x$$

The next steps are the 12 attention and feed-forward layers. Attention looks at the relationship of a word to the other words and captures the nuances of text to improve performance. For example, it is able to capture the different meanings of bear in “I can’t bear it” and “A brown bear”.

Attention is defined by the following. The query Q , key K , and value V matrices are calculated using their respective weight matrices W_Q, W_K, W_V . Furthermore, d_k is the dimension of a key vector k_i . The attention layers are defined in the following:

For layer $l = 1, 2, \dots, 12$

$$h^{l+0.5} = h^l + \text{Attention}(h^l)$$
$$\text{Attention}(h^l) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$
$$q_t = W_Q^l h_t^l, \quad k_i = W_K^l h_i^l, \quad v_i = W_V^l h_i^l$$

$$Q = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}, K = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{bmatrix}, V = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

After attention is calculated, the results are fed into a feed-forward neural network that has 2 layers where W_1 and b_1 are the weights and biases for the first layer while W_2 and b_2 are for the second. The feed-forward layers look like:

$$h^{l+1} = h^{l+0.5} + \text{MLP}(h^{l+0.5})$$

$$\text{MLP}(h^{l+1}) = W_2^l \sigma(W_1^l h^{l+0.5} + b_1^l) + b_2^l$$

The results will then be unembedded with unembed matrix W_u to get a logit score s . Then, since this is a binary classification problem, a sigmoid activation is applied to get a probability p . The equations for s and p are defined in the following:

$$s = W_u h$$

$$p = \sigma(s)$$

The model was fine-tuned using Binary Cross-Entropy loss with the Adam optimizer. A batch size of 16 was used, and training was limited to one epoch, as additional epochs resulted in overfitting. The learning rate was set to $2 * 10^{-5}$, and an L_2 regularization penalty α of 0.001 was applied.

CHAPTER 6

Evaluation

Before discussion can begin regarding how well the models did, we must specify which metrics are being used and their interpretations.

- **True Positive Rate (TPR):** The proportion of depressed observations correctly classified as depressed
- **True Negative Rate (TNR):** The proportion of non-depressed observations correctly classified as non-depressed
- **Accuracy:** The overall proportion of correctly classified individuals

The top priority is to select a model with a high true positive rate (TPR), as false negatives, cases in which depressed individuals are classified as non-depressed, are particularly problematic. Such errors could delay treatment and potentially place the individual's health at risk. The true negative rate (TNR) will be considered a secondary metric, as false positives are generally less harmful in this context. In the best case, a healthy individual may simply disregard the result; in the worst case, they may seek professional evaluation to confirm the outcome. Nevertheless, it is still desirable to keep the false positive rate relatively low in order to minimize unnecessary concern and avoid the diversion of limited healthcare resources.

6.1 Classical Models Performance

6.1.1 k-Nearest Neighbors

The results of k-NN are as follows

Accuracy	TPR	TNR
72.8%	76.7%	68.8%

Table 6.1: Metrics for k-NN

k-NN had the worst overall performance among all models since it had the lowest TPR at 76.7%. Furthermore, it struggled to classify non-depressed observations properly due to how low its TNR was at 68.8%. Additionally, this model also performs poorly when it comes to computational costs. While it does not do any training, there is a constant need to calculate the cosine distances with the training data for the testing data. This can get expensive if the data is very large.

6.1.2 L_2 Logistic Regression

Accuracy	TPR	TNR
76.2%	85.8%	65.9%

Table 6.2: Metrics for L_2 Logistic Regression

Meanwhile, logistic regression performed noticeably better for TPR at 85.8% but did slightly worse for the TNR at 65.9%. While optimizing the loss function is slightly costly, it is inexpensive to make an evaluation with the testing data once the coefficients have been estimated. Additionally, some added benefits of logistic regression is the interpretability of the coefficients and to identify which individual words had an effect on the odds. For

example, the coefficient for a word x , β can be interpreted as the increase in a unit for word x increases/decreases the odds of depression by e^β .

6.1.3 Multinomial Naive Bayes

Accuracy	TPR	TNR
79.8%	88.7%	70.3%

Table 6.3: Metrics for Multinomial Naive Bayes

Among all the classical models, Multinomial Naive Bayes performed the best in terms of the TPR and TNR at 88.7% and 70.3% respectively. It does exceptionally well at classifying depressed people correctly. Additionally, this model has the cheapest computational costs. The model simply needs to compute the counts and probabilities (which is fairly straightforward) one time to train itself and make evaluations.

6.2 Deep Learning Models Performance

6.2.1 Feed-Forward Neural Network

Accuracy	TPR	TNR
81.1%	79.2%	83.2%

Table 6.4: Metrics for Feed-Forward Neural Network

Despite the large complexity gap for neural networks and classical models, the difference in overall performance was not that big. The feed-forward neural network had an accuracy of 81.1% which is not that different from the best classical model Multinomial Naive Bayes at 79.8%. However, one notable finding is that the feed-forward network had a much higher TNR at 83.2% and does really well at classifying non-depressed people. While its TPR is

relatively high at 79.2%, there are still models that beat it in this area (which is the top priority).

Circling back to one of our research questions, it raises the question of whether it was worth all the extra resources. The neural network was significantly more costly as it has a complex architecture and required parallel computing to train in a reasonable time, yet it was beat out by a cheaper classical model Multinomial Naive Bayes in terms of the TPR.

6.2.2 Fine-Tuned BERT

Accuracy	TPR	TNR
85.2%	86.2%	84.2%

Table 6.5: Metrics for BERT

BERT overall has the most balanced performance among all the models. It has both relatively high TPR and TNR at 86.2% and 84.2% and seems to classify non-depressed and depressed people very well. However, the costs of this model were the highest. The architecture is the most complex, and a GPU was required to train the model.

6.3 Summary

As a recap of all the models, the table below summarizes all their metrics. Discussion will be expanded upon in the next chapter.

Model	Accuracy	TPR	TNR
k-NN	72.8%	76.7%	68.8%
Logistic Regression	76.2%	85.8%	65.9%
Multinomial NB	79.8%	88.7%	70.3%
Feed-Forward NN	81.1%	79.2%	83.2%
BERT	85.2%	86.2%	84.2%

Table 6.6: Comparison of Metrics Between All Models

CHAPTER 7

Discussion & Conclusion

7.1 Limitations

7.1.1 Data Imperfections

Although the overall sentiment of comments from r/depression and r/AskReddit generally aligns with the labels of depressed and non-depressed, this correspondence is not perfect. Posting a comment in r/depression does not necessarily mean that the individual is clinically depressed. Similarly, a user posting in r/AskReddit may still be experiencing depression without it being apparent in the data. Therefore, a more accurate interpretation of the labels is that a value of 1 indicates that the comment originated from r/depression, while a value of 0 indicates that the comment originated from r/AskReddit. Despite this limitation, comments from r/depression remain useful for identifying depressive language patterns, as they are strongly correlated with discussions related to depression.

7.1.2 Formal Diagnosis

These models should not be used to make formal clinical diagnoses, as they are not trained medical professionals. Instead, they are more appropriately used as screening tools to flag individuals who may be experiencing symptoms of depression. Such systems could encourage individuals to seek professional evaluation or support when appropriate.

7.2 Future Work

7.2.1 Time Variable

A point of interest would be exploring the effects of time on depression. Our current models only take word count into consideration, and future work could attempt to incorporate time as a variable. Since the data spans from 2005-2024, it would be interesting to see how the comments from the pre-COVID, COVID, and post-COVID eras differ in terms of depression rates. It also might be useful to observe how the vocabulary associated with depression changed with time.

7.2.2 Language Expansion

Because the models only utilized English text, an important direction for future work would be the development of models capable of handling multiple languages. English-speaking populations represent only a portion of the global population, and expanding language coverage could allow such systems to assist individuals in a wider range of countries and communities.

7.2.3 Additional Data

The dataset used in this analysis did not include the complete available data, as the approximately 90GB dataset from r/AskReddit exceeded the computational resources available for this study. Future research could attempt to incorporate the full dataset, which may improve model performance by providing additional training examples. Additionally, there are other subreddits and social media sites that contain discussions from both depressed and non-depressed individuals, which could further enrich the dataset.

7.2.4 Additional Disorders

This study focused on binary classification between depression and non-depression. Future work could extend this framework to a multi-class classification problem involving multiple mental health conditions, such as anxiety or bipolar disorder. For example, data could be collected from communities such as r/anxiety and r/bipolar to then train models capable of distinguishing between different types of mental health conditions.

7.2.5 Additional Models

This study examined only a subset of possible models. Numerous additional classical machine learning methods and transformer-based architectures could also be explored. In particular, different variations and improvements of BERT-based models may provide further performance gains.

7.3 Conclusion

The primary objective of this study was to compare classical machine learning models with deep learning models in terms of predictive performance and to evaluate whether any performance improvements justified the additional computational costs. Returning to our research questions, among the classical models, Multinomial Naive Bayes performed the best and achieved a TPR of 88.7%, TNR of 70.3%, and an overall accuracy of 79.8%. The best overall model was BERT, which achieved a TPR of 86.2%, a TNR of 84.2%, and an accuracy of 85.2%.

The performance gap between the best classical model and the deep learning model was primarily observed in TNR and overall accuracy, with differences of approximately 14% and 5%, respectively. The BERT model performed particularly well at correctly classifying non-depressed individuals, an area where Multinomial Naive Bayes struggled. However, because

false positives are generally less problematic than false negatives in this context, the relatively lower TNR of Multinomial Naive Bayes should not be overly penalized. In fact, Multinomial Naive Bayes achieved a 2.5% higher TPR than BERT, which is arguably the most critical metric in this task. Given that the model maintained a reasonably acceptable TNR, its performance remains competitive.

Since the primary difference between the models lies in TNR, the decision ultimately depends on whether the improvement in correctly identifying non-depressed individuals justifies the additional computational cost of training and deploying a transformer model. Because this system is intended to flag potential cases of depression rather than provide a formal diagnosis, errors are less consequential unless they occur for the depressed class. Furthermore, Multinomial Naive Bayes is computationally inexpensive and performs slightly better in terms of TPR compared to BERT. Therefore, Multinomial Naive Bayes may represent a practical and efficient alternative to BERT, particularly in settings where computational resources required to train transformer-based models are limited.

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